

EXPRESS CAR - Full Application Viva Report (Gujarati)

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Project: Express Car (Flutter + Firebase)

1) Project Overview

Express Car એક Flutter based car rental application છે, જેમાં user side, owner/business side અને admin side એમ ત્રણ મુખ્ય role-based flow છે.

આ app માં મુખ્ય રીતે નીચેના feature set છે:

- User authentication (Email/Password + Google Sign-In)
- Profile completion અને role-based routing
- Car listing, category filtering, favorites
- Car booking, booking details, booking history
- Live location tracking with map
- Owner side car/fleet management
- Admin side users/fleet/driver document monitoring

2) Tech Stack

- Frontend: Flutter (Dart)
- Backend: Firebase (Auth, Firestore, Storage)
- External APIs/Services: Google Sign-In, ImgBB API, Cloudinary API, Geolocator
- Mapping: flutter_map + latlong2

3) Libraries Used (pubspec + code usage)

1. flutter
2. firebase_core
3. firebase_auth
4. cloud_firestore
5. firebase_storage
6. google_sign_in
7. pininput
8. http
9. image_picker
10. multi_select_flutter
11. flutter_launcher_icons
12. cupertino_icons
13. cloudinary_public
14. geolocator
15. flutter_map
16. latlong2

Dev dependencies:

- flutter_test
- flutter_lints

4) Application Startup & Routing Flow

1. `main.dart`

- Firebase initialize થાય છે.
- Initial car data fetch થાય છે (`fetchCarsFromFirestore`).
- `SplashScreenWrapper` દ્વારા splash પછી `AuthWrapper` પર navigation.

2. `Authentication/auth\_wrapper.dart`

- `FirebaseAuth.authStateChanges()` stream સંભળે છે.
- User login state check કરે છે.
- Firestore `users` document પરથી role (`admin` / `user`) resolve કરે છે.
- Profile complete છે કે નહીં તે ચેક કરે છે.
- Final routing:
 - Admin -> `AdminPanelPage`
 - Incomplete profile -> `CompleteProfilePage`
 - Normal user -> `HomePage`

5) Complete File-by-File Purpose (lib folder)

Total Dart files: 44

A) Root / Core

1. `lib/main.dart` - App entrypoint, Firebase init, splash-to-auth flow.
2. `lib/firebase\_options.dart` - Platform-wise Firebase config values.
3. `lib/theme/app\_theme.dart` - Global app theme tokens/colors/styles.

B) Splash Module

4. `lib/Splash/splash.dart` - Splash screen UI.
5. `lib/Splash/get\_start.dart` - Get Started style landing/pre-auth info page.

C) Authentication Module

6. `lib/Authentication/auth\_wrapper.dart` - Auth state + role-based routing.
7. `lib/Authentication/signup\_page.dart` - New user registration.
8. `lib/Authentication/signin\_page.dart` - Main sign-in page (email/password + Google).
9. `lib/Authentication/signin\_page\_new.dart` - Alternate sign-in implementation (legacy/duplicate style).
10. `lib/Authentication/admin\_login\_page.dart` - Dedicated admin login path.
11. `lib/Authentication/verify\_email\_page.dart` - Email verification flow.

- 12. `lib/Authentication/CompleteProfile.dart` - Mandatory profile details collection.
- 13. `lib/Authentication/enter\_email.dart` - Password reset email input.
- 14. `lib/Authentication/enter\_otp.dart` - OTP verification page.
- 15. `lib/Authentication/enter\_otp\_repass.dart` - OTP + reset progression page.
- 16. `lib/Authentication/enter\_newpassword.dart` - Password update completion.
- 17. `lib/Authentication/TermsConditions.dart` - Terms and conditions page.

D) Home / Cars / Favorites Module

- 18. `lib/HomeDetails/Home\_Page/home\_page.dart` - Main dashboard, categories, search, favorites sync.
- 19. `lib/HomeDetails/Home\_Page/car\_model.dart` - Car model/data class.
- 20. `lib/HomeDetails/Home\_Page/car\_data.dart` - Local fallback cars + Firestore fetch merge logic.
- 21. `lib/HomeDetails/Favorite\_car/Favorite.dart` - Favorited car list UI.

E) Booking Module

- 22. `lib/HomeDetails/Booking/Book\_car.dart` - Booking form, date/unit/driver details capture.
- 23. `lib/HomeDetails/Booking/booking\_details\_page.dart` - Booking confirmation/detail summary + rating handling.
- 24. `lib/HomeDetails/Booking/Booked\_Car.dart` - User booking history/active bookings view.
- 25. `lib/HomeDetails/Booking/live\_booking\_tracking\_page.dart` - Live GPS tracking + map display.

F) Menu & Profile Module

- 26. `lib/HomeDetails/Menu/Menu.dart` - Menu drawer/page, profile summary, logout actions.
- 27. `lib/HomeDetails/Menu/Menus\_Files/Account.dart` - Account page.
- 28. `lib/HomeDetails/Menu/Menus\_Files/ViewProfile.dart` - Profile read-only view.
- 29. `lib/HomeDetails/Menu/Menus\_Files/EditProfile.dart` - Profile edit + image update.
- 30. `lib/HomeDetails/Menu/Menus\_Files/ChangePassword.dart` - Password change flow.

G) Owner / Business Handling Module

- 31. `lib/Handle\_Car/HandleBussiness.dart` - Owner/business navigation hub.
- 32. `lib/Handle\_Car/DashBoard.dart` - Business stats/dashboard.
- 33. `lib/Handle\_Car/Add\_Car.dart` - Add car form + image/source handling + save.
- 34. `lib/Handle\_Car/Manage\_Cars.dart` - Car management (list/update/delete style operations).
- 35. `lib/Handle\_Car/Manage\_Booking.dart` - Booking management for owner side.
- 36. `lib/Handle\_Car/Done.dart` - Success/confirmation page after operation.

H) Admin Module

- 37. `lib/Admin/admin\_panel.dart` - Admin dashboard with aggregate stats and monitoring cards.
- 38. `lib/Admin/manage\_users.dart` - Users list + search + view.
- 39. `lib/Admin/fleet\_driver\_manager.dart` - Fleet + driver document workflow management.

I) Services Module

- 40. `lib/services/user\_presence\_service.dart` - Online/offline presence and timestamp updates.
- 41. `lib/services/car\_location\_tracking\_service.dart` - Real-time location stream and Firestore updates.
- 42. `lib/services/imgbb\_upload\_service.dart` - ImgBB multipart upload helper.
- 43. `lib/services/car\_image\_widget.dart` - Reusable widget for car image rendering.

J) Extra User Page

- 44. `lib/UserPages/dashboard\_page.dart` - Additional dashboard/user navigation style page.

6) APIs / Services Used and Why

6.1 Firebase Authentication API

Use cases:

- Sign up, sign in, sign out
- Auth state listener
- Password reset
- Email verification
- Google credential-based sign-in

Typical methods used:

- `createUserWithEmailAndPassword`
- `signInWithEmailAndPassword`
- `signInWithCredential`
- `authStateChanges`
- `sendPasswordResetEmail`
- `currentUser`, `reload`, `sendEmailVerification`

6.2 Cloud Firestore API

Primary database for all app entities.

Main collections found in code:

- `users`
- `cars`
- `bookings`
- `fleet\_items`
- `driver\_documents`

Use cases:

- Role storage (`users.role`)
- Profile fields save/update
- Car inventory read/write
- Booking creation/history/status
- Favorites persistence
- Presence tracking (`isOnline`, `lastSeenAt`, `lastLoginAt`)
- Live location related geo fields

6.3 Firebase Storage API

Use case:

- Profile image upload/store (with fallback strategy in some flows)

6.4 Google Sign-In API

Use case:

- Social login for user convenience
- OAuth token -> Firebase credential link

6.5 ImgBB HTTP API

Use case:

- Image upload fallback/alternative pipeline
- Multipart HTTP upload via `http` package

6.6 Cloudinary API

Use case:

- Fleet/driver document uploads in admin driver manager module

6.7 Geolocator API

Use case:

- Permission checks
- Current position અને continuous position stream
- Booking/live tracking and car location updates

6.8 Flutter Map + LatLong2

Use case:

- Live booking tracking map UI rendering
- User/car positions visualized on map

7) Functional Flow for Viva Explanation

User Flow

1. Splash -> AuthWrapper
2. Sign In / Sign Up
3. Email verify + profile complete
4. Home page -> browse cars
5. Book car -> booking details -> booked list
6. Optional live tracking on map
7. Menu -> view/edit profile, change password, logout

Owner/Business Flow

1. Login as normal account (owner-type data)
2. Handle Business page
3. Add Car
4. Manage Cars
5. Manage Bookings
6. Dashboard stats

Admin Flow

1. Admin login / role-based admin routing
2. Admin panel aggregate metrics
3. Manage users
4. Fleet + driver documents handling

8) Data Model Summary (Conceptual)

- User: uid, name, email, phone, address, dob, gender, role, profile image, online status
- Car/Fleet: name, model, type, price, location, images, owner info, availability, geo info
- Booking: user info, car reference, start/end date, rental unit, pickup details, status, amounts, optional docs/ratings

9) Security & Validation Notes (Viva Points)

- Role-based gatekeeping via AuthWrapper + Firestore role field.
- Email verification flow integrated.
- Presence and activity timestamps maintained.
- Booking and owner/admin operations separated by page/module.
- External uploads handled via dedicated service files.

10) Potential Improvement Points (Good Viva Discussion)

1. Duplicate sign-in page (`signin\_page.dart` and `signin\_page\_new.dart`) cleanup.
2. Strong typed repository/service layer introduce *දැඩි* UI-logic coupling *අවම*.

3. Firestore security rules role-based hardening.
4. Automated tests (unit/widget/integration) વધારવા.
5. Booking/payment abstraction if payment gateway integration planned.

11) Quick Viva One-Liner

"Express Car is a role-based Flutter car rental platform using Firebase Auth + Firestore as core backend, with booking, live tracking, owner fleet management, and admin monitoring modules, plus hybrid media upload support through Firebase Storage, ImgBB, and Cloudinary integrations."

12) File Count & Coverage Summary

- Total Dart files in lib: 44
- Major modules: Authentication, Home/Booking, Menu/Profile, Handle_Car (owner), Admin, Services
- Core APIs: Firebase (Auth/Firestore/Storage), Google Sign-In, HTTP(ImgBB), Cloudinary, Geolocator, FlutterMap

આ document Viva માટે full application explanation તરીકે તૈયાર છે.